

FORM 5 CHEMISTRY

CHAPTER 5
CONSUMER AND
INDUSTRIAL CHEMISTRY
消費者與工業化學

Name:

Class:

Teacher:

Date:

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How to Use This Workbook / 使用方式

During the lesson, listen for the missing English keywords and complete every underlined space. Chinese translations are provided beside selected scientific terms to support understanding, but students should practise writing the English terms. After completing the guided notes, study the worked examples and answer the practice questions without referring to the notes.

Teacher note: Missing words are printed in blue, and model answers are included for discussion.

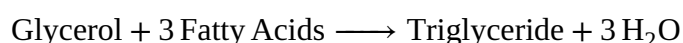
5.1 Oils and Fats / 油與脂肪

Suggested lesson focus: structure, saturation, physical properties and hydrogenation

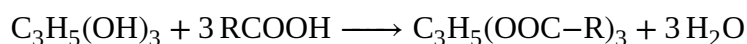
A. Formation of Oils and Fats / 油與脂肪的形成

Oils and fats are naturally occurring **esters** (酯類). Most oils and fats are **triglycerides** (三酸甘油酯).

A triglyceride is formed when one **glycerol** (甘油) molecule reacts with three **fatty acid** (脂肪酸) molecules.



A simplified general equation is:



In this equation, *R* represents a long **hydrocarbon chain** (碳氫鏈). The reaction is called **esterification** (酯化反應) because an alcohol reacts with a carboxylic acid to form an ester and water.

Key Vocabulary / 關鍵詞彙

Glycerol is also called **propan-1,2,3-triol** (丙烷-1,2,3-三醇). It contains three **hydroxyl groups** (羥基), so one glycerol molecule can form three **ester linkages** (酯鍵).

B. Saturated and Unsaturated Fatty Acids / 飽和與不飽和脂肪酸

Aspect	Saturated Fatty Acid (飽和脂肪酸)	Unsaturated Fatty Acid (不飽和脂肪酸)
Carbon-carbon bonds	Contains only single bonds (單鍵)	Contains at least one double bond (雙鍵)
Shape of chain	Relatively straight	Bent or kinked (彎折的) near a cis double bond
Packing of molecules	Molecules pack more closely	Molecules do not pack as closely
Intermolecular attraction	Generally stronger (較強)	Generally weaker (較弱)
Melting point	Generally higher	Generally lower
Typical state at room temperature	More likely to be solid	More likely to be liquid
Examples	Palmitic acid, stearic acid	Oleic acid, linoleic acid

C. Comparison Between Oils and Fats / 油與脂肪的比較

Aspect	Oils (油)	Fats (脂肪)
Physical state at room temperature	Usually liquid	Usually solid or semi-solid
Main fatty acid content	Higher percentage of unsaturated fatty acids (不飽和脂肪酸)	Higher percentage of saturated fatty acids (飽和脂肪酸)
Melting point	Generally lower	Generally higher
Common examples	Olive oil, soybean oil, corn oil	Butter, ghee, animal fat

Common Mistake

Do not assume that every plant product is an oil or every animal product is a fat. Palm oil and coconut oil are plant products but contain relatively high proportions of **saturated fatty acids** (飽和脂肪酸). Fish oils are animal products but can contain high proportions of **unsaturated fatty acids** (不飽和脂肪酸).



Visual Link / 圖像連結: The vegetable oil remains liquid because it generally contains a higher proportion of **unsaturated fatty acids**, while butter is more solid because it contains more **saturated fatty acids**.

Visual Check / 看圖填空

Oil usually has a **lower melting point**, whereas fat usually has a **higher melting point**. The difference is related to molecular shape, packing and **intermolecular attractions** (分子間作用力).

D. Why Oils Are Usually Liquid / 為何油通常是液體

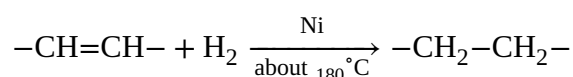
Unsaturated fatty acid chains contain carbon-carbon **double bonds** (雙鍵). A **cis double bond** (順式雙鍵) produces a bend in the chain.

The bent molecules cannot **pack closely**, so the **intermolecular attractions** (分子間作用力) are weaker. Less **thermal energy** (熱能) is required to separate the molecules, giving a lower melting point.

Saturated fatty acid chains are **straighter** and can pack closely. Their intermolecular attractions are stronger, so more thermal energy is needed and their melting points are **higher**.

E. Hydrogenation / 氫化反應

Hydrogenation is an **addition reaction** (加成反應) in which hydrogen is added across carbon-carbon double bonds in an unsaturated oil.



Conditions / 條件:

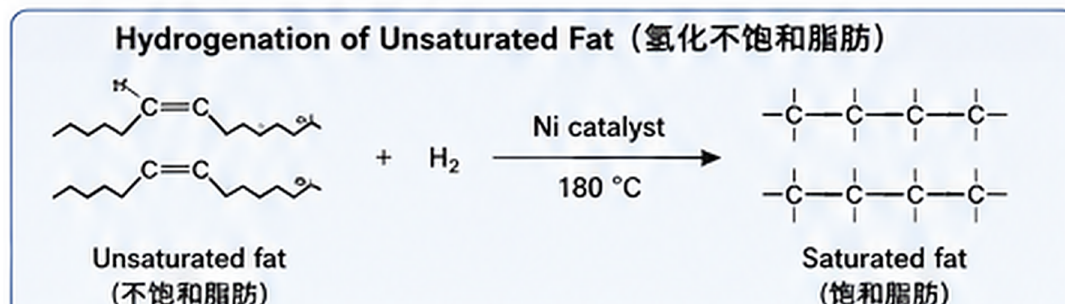
- **hydrogen gas** (氫氣), H_2 ;
- a **nickel catalyst** (鎳催化劑);
- heating to approximately 180°C .

During hydrogenation:

1. the number of carbon-carbon **double bonds** decreases;

- the degree of **saturation** (飽和程度) increases;
- the molecules become **straighter** and pack more closely;
- the intermolecular attractions become **stronger**;
- the melting point **increases**;
- the product becomes more **solid** at room temperature.

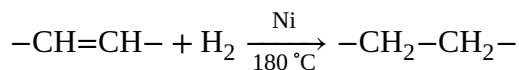
Hydrogenation is used industrially to convert **liquid vegetable oils** (液態植物油) into semi-solid products such as **margarine** (人造奶油).



Visual Link / 圖像連結: Hydrogen adds across each carbon-carbon **double bond**. The product contains more carbon-carbon **single bonds** and is therefore more saturated.

Read the Diagram / 看圖理解

Complete the change:



The nickel acts as a **catalyst** (催化劑) and is not consumed in the reaction.

Worked Example

Question: Vegetable oil is reacted with hydrogen at 180°C in the presence of nickel. Explain why the product is more solid at room temperature.

Model answer: Hydrogen adds across the carbon-carbon **double bonds**. The oil becomes more **saturated**, and the hydrocarbon chains become straighter. The molecules can pack more closely, producing stronger **intermolecular attractions** (分子間作用力). Therefore, the melting point increases and the product becomes more solid at room temperature.

F. Quick Check

1. State the type of compound to which oils and fats belong. [1 mark]

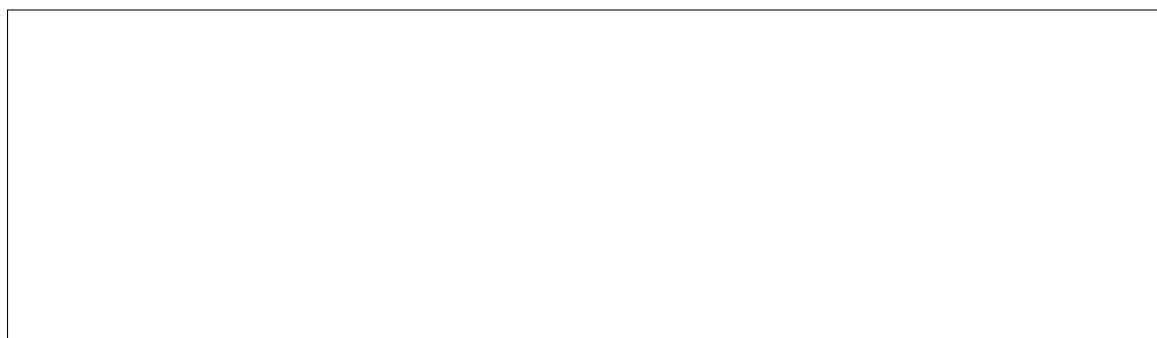
2. Name the alcohol used to form triglycerides. [1 mark]

3. State one structural difference between a saturated fatty acid and an unsaturated fatty acid. [2 marks]

4. Explain why an unsaturated oil usually has a lower melting point than a saturated fat. [3 marks]

5. State the reagent, catalyst and temperature used in hydrogenation. [3 marks]

6. Draw a simple structural representation to show what happens to one carbon-carbon double bond during hydrogenation. [2 marks]



5.2 Cleaning Agents / 清潔劑

Suggested lesson focus: soap preparation, detergent structure, micelles and hard water

A. Soap / 肥皂

Soap is a sodium or potassium **salt** (鹽) of a long-chain fatty acid. The general formula of a soap anion is:

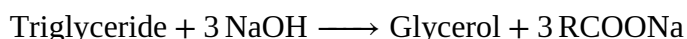


Examples include **sodium stearate** (硬脂酸鈉), $\text{C}_{17}\text{H}_{35}\text{COONa}$, and **potassium palmitate** (棕櫚酸鉀), $\text{C}_{15}\text{H}_{31}\text{COOK}$.

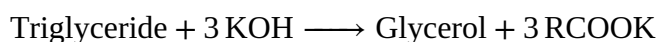
Sodium soaps are usually **harder** and are commonly used in solid soap bars. Potassium soaps are usually **softer** and are commonly used in liquid or soft soaps.

B. Saponification / 皂化反應

Saponification is the **alkaline hydrolysis** (鹼性水解) of an oil or fat to produce **soap** and **glycerol** (甘油).



If potassium hydroxide is used, **potassium soap** (鉀皂) is formed:



Laboratory Preparation of Soap / 實驗室製皂

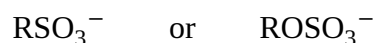
1. Mix vegetable oil with **concentrated sodium hydroxide** (濃氫氧化鈉) solution.
2. **Heat** the mixture while stirring carefully.
3. Add **saturated sodium chloride** (飽和氯化鈉) solution.
4. Allow the mixture to **cool**.
5. **Filter** the solid soap.
6. Wash the soap with a small amount of cold **distilled water** (蒸餾水).
7. **Press and dry** the soap.

Function of Sodium Chloride / 氯化鈉的作用

Sodium chloride **reduces the solubility** of soap in water, causing the soap to separate as a solid. This process is called **salting out** (鹽析).

C. Detergents / 合成清潔劑

Detergents are **synthetic cleaning agents** (合成清潔劑). Common detergent anions are **alkylbenzene sulphonates** (烷基苯磺酸鹽) or **alkyl sulphates** (烷基硫酸鹽).



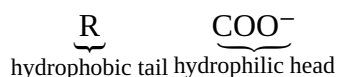
Detergents are commonly manufactured from **petroleum-derived raw materials** (石油衍生原料).

D. Structure of Soap and Detergent Anions / 肥皂與清潔劑陰離子的結構

Both soap and detergent anions contain two important parts:

- a **hydrophobic hydrocarbon tail** (疏水性碳氫尾部), which is attracted to oil and grease but not to water;
- a **hydrophilic ionic head** (親水性離子頭部), which is attracted to water.

For a soap anion:



E. Cleansing Action / 清潔作用

The cleansing action occurs in stages:

1. Soap or detergent lowers the **surface tension** (表面張力) of water, allowing water to wet the fabric more effectively.
2. The hydrophobic tails **dissolve in the grease**.
3. The hydrophilic heads remain in the surrounding **water**.
4. **Agitation** (攪動) breaks the grease into smaller droplets.
5. Soap or detergent anions surround each grease droplet to form a **micelle** (膠束).
6. The similarly charged heads **repel** one another, preventing the droplets from joining again.
7. The suspended grease droplets are washed away during **rinsing** (沖洗).

Formation of Micelle (膠束的形成)

Visual Link / 圖像連結: Agitation breaks grease into small droplets. In a micelle, the **hydrophobic tails** (疏水尾部) point towards the grease, while the **hydrophilic heads** (親水頭部) face the surrounding water.

Micelle Orientation / 膠束方向

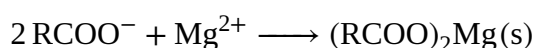
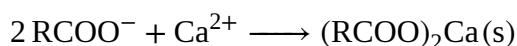
Inside the micelle, the hydrocarbon tails point **inwards** and the ionic heads point **outwards**. The charged heads repel one another and help keep the grease droplets **suspended in water**.

Exam Tip

For a 4–6 mark cleansing-action question, mention the hydrophobic tail, hydrophilic head, agitation, micelle formation and rinsing. Merely writing “soap removes grease” is not enough.

F. Soap in Hard Water / 肥皂在硬水中

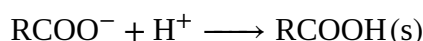
Hard water contains **calcium ions** (鈣離子), Ca^{2+} , and **magnesium ions** (鎂離子), Mg^{2+} . Soap anions react with these ions to form **insoluble salts** (不溶性鹽) called **scum** (浮渣/皂垢).



Scum wastes soap, reduces **lather formation** (泡沫形成) and can leave a deposit on clothes or containers. Detergents generally do not form insoluble precipitates with calcium and magnesium ions, so they remain **effective** in hard water.

G. Soap in Acidic Water / 肥皂在酸性水中

Soap anions react with **hydrogen ions** (氫離子) to form insoluble fatty acids:



Therefore, soap is **less effective** in acidic water. Detergents generally remain effective under **acidic conditions** (酸性條件).

H. Comparison of Soap and Detergent / 肥皂與清潔劑的比較

Aspect	Soap (肥皂)	Detergent (合成清潔劑)
Raw material	Oils and fats (油與脂肪)	Usually petroleum-derived raw materials
Functional group	Carboxylate (羧酸根), $-\text{COO}^-$	Sulphonate or sulphate (磺酸根或硫酸根)
Soft water	Effective	Effective
Hard water	Forms scum and becomes less effective	Does not normally form scum; remains effective
Acidic water	Forms an insoluble fatty acid	Generally remains effective
Biodegradability	Usually readily biodegradable (可生物分解)	Depends on molecular structure

Worked Example

Question: Explain why detergent is more effective than soap in hard water.

Model answer: Hard water contains Ca^{2+} and Mg^{2+} ions. Soap anions react with these ions to form **insoluble salts** called scum. This reduces the amount of soap available for cleansing. Detergent anions do not normally form insoluble **precipitates** (沉澱) with these ions, so detergent remains effective.

I. Quick Check

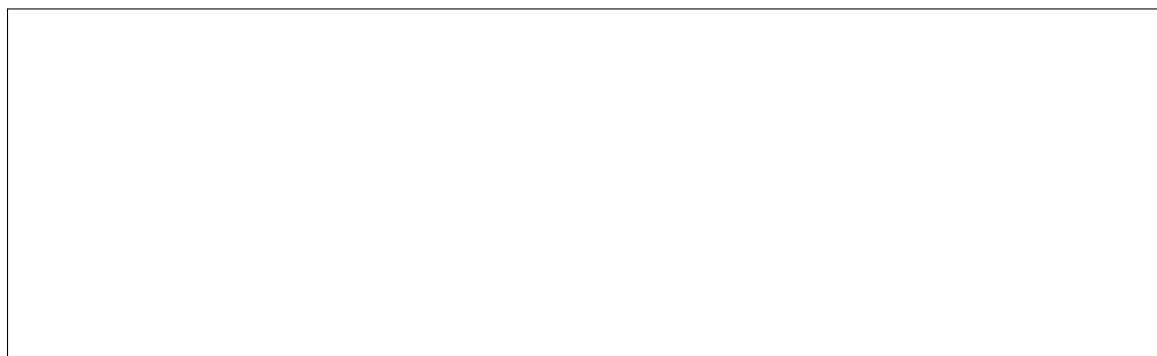
1. Define saponification. [2 marks]

2. State the two products of saponification. [2 marks]

3. Explain the function of saturated sodium chloride solution during soap preparation. [2 marks]

4. State the two parts of a soap anion and the substance to which each part is attracted. [4 marks]

5. Draw a labelled micelle surrounding a grease droplet. Show the correct orientation of the soap or detergent particles. [4 marks]



6. Write an ionic equation for the formation of calcium soap scum. [2 marks]

7. Explain why soap is less effective in acidic water. [2 marks]

5.3 Food Additives / 食品添加劑

Suggested lesson focus: functions, examples, food labels and responsible use

A. Definition / 定義

Food additives are substances **deliberately added** (刻意加入) to food to preserve it or improve its taste, smell, appearance, texture or **stability** (穩定性) .

B. Main Types of Food Additives / 主要種類

Type	Function	Examples
Preservative (防腐劑)	Prevents or slows microbial growth and delays spoilage	Salt, sugar, vinegar, sodium benzoate, sodium nitrite, sulphur dioxide
Antioxidant (抗氧化劑)	Slows oxidation , especially of fats and oils, and delays rancidity	Ascorbic acid, tocopherol, BHA, BHT
Flavouring agent (調味劑)	Improves or strengthens taste and smell	Monosodium glutamate, vanilla essence, fruit-flavoured esters
Colouring agent (著色劑)	Restores or improves colour and appearance	Caramel, chlorophyll, turmeric, permitted synthetic dyes
Emulsifier (乳化劑)	Helps two immiscible liquids (互不相溶液體) form an emulsion	Lecithin, mono- and diglycerides
Stabiliser (穩定劑)	Maintains a uniform mixture and prevents separation	Lecithin, gelatin, gums
Thickener (增稠劑)	Increases viscosity (黏度) or forms a gel-like texture	Pectin, gelatin, starch, xanthan gum

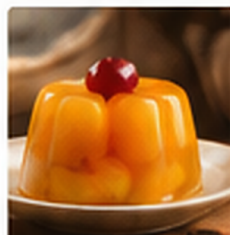
C. Important Examples / 重要例子

Ascorbic acid (抗壞血酸／維生素 C) : It is an **antioxidant**. It is oxidised more readily than many food components, so it reacts with oxygen first and **delays oxidation** of the food.

Lecithin (卵磷脂) : It is an **emulsifier and stabiliser**. It contains regions that interact with water and oil, helping an oil-water mixture remain **uniform**.

Pectin (果膠) : It is a **thickener** that absorbs water and forms a **gel** (凝膠) . It is commonly used in jams and fruit products.

Sodium benzoate (苯甲酸鈉) : It is a **preservative** used especially in acidic foods and drinks to inhibit **microbial growth** (微生物生長) .



Visual Link / 圖像連結: Different foods require different additives. Pickled foods often use **preservatives**; jelly uses a **thickener**; and ice cream commonly uses emulsifiers and stabilisers to maintain a smooth texture.

Food Product Matching / 食品配對

Match each product to the most likely function:

- Pickled vegetables: slowing **microbial growth**
- Jelly: increasing **viscosity** (黏度) and forming a gel
- Ice cream: preventing oil and water phases from **separating**

D. Responsible Use / 安全使用

Food additives must be used within **permitted limits** (允許限量). Excessive intake may cause undesirable effects in **sensitive individuals** (敏感人士). Food additives must not be used to hide spoilage, contamination or poor-quality ingredients.

Exam Tip

When a food label is given, do not only name the additive type. Link the additive to the actual product. Example: "Lecithin acts as an emulsifier to keep the oil and water phases of mayonnaise mixed."

E. Quick Check

A salad dressing contains lecithin, ascorbic acid, sodium benzoate and permitted colouring.

1. State the type and function of lecithin. [2 marks]

2. State the type and function of ascorbic acid. [2 marks]

3. State the type and function of sodium benzoate. [2 marks]

4. Explain why an emulsifier is important in salad dressing. [2 marks]

5. Jam contains pectin. State the role of pectin and explain the change it causes in the jam. [2 marks]

6. Suggest one reason why food additives are regulated. [1 mark]

5.4 Medicines and Cosmetics / 藥物與化妝品

Suggested lesson focus: functions, correct use, side effects and consumer safety

A. Medicines / 藥物

Medicines are chemicals used to **prevent, control or treat** diseases and their symptoms. Medicines may be classified as **traditional medicines** (傳統藥物) or **modern medicines** (現代藥物).

B. Traditional Medicines / 傳統藥物

Traditional medicines are mainly obtained from **natural sources** (天然來源) such as plants. They may be prepared as extracts, powders, liquids, tablets or capsules.

Natural origin does not automatically mean **safe**. Dosage, purity, interactions and **contamination** (污染) must still be considered.

Source	Common Traditional Use
Aloe vera (蘆薈)	Relieves minor skin irritation and discomfort from mild burns
Ginger (薑)	Relieves nausea and digestive discomfort
Garlic (大蒜)	Traditionally consumed to support general health
Turmeric (薑黃)	Traditionally used for minor skin conditions and inflammation
Ginseng (人參)	Traditionally consumed to improve energy and general well-being

C. Analgesics / 止痛藥

Analgesics (止痛藥) relieve pain without necessarily curing the underlying disease.

Aspirin (阿司匹林) : It relieves pain, fever and **inflammation**. It may irritate the **stomach lining** (胃壁) and increase the risk of bleeding. It is generally unsuitable for patients with gastric ulcers and should not be given to children or teenagers with viral infections unless directed by a doctor.

Paracetamol (撲熱息痛) : It relieves pain and reduces **fever**. It generally causes less stomach irritation than aspirin, but an **overdose** (過量服用) may cause severe liver damage.

Codeine (可待因) : It is a stronger analgesic that may cause drowsiness, constipation, dependence or **addiction**. It must be used under medical supervision.

D. Antibiotics / 抗生素

Antibiotics (抗生素) kill bacteria or inhibit their growth. Examples include penicillin and streptomycin. Antibiotics treat **bacterial infections**, not viral infections such as influenza or the common cold.

A patient must complete the prescribed antibiotic course because:

1. stopping early may leave some bacteria **alive**;
2. **less susceptible bacteria** (較不敏感的細菌) are more likely to survive;

- the surviving bacteria **reproduce**;
- the proportion of **resistant bacteria** (抗藥性細菌) increases;
- the infection may return and become more difficult to **treat**.

Common Mistake

Do not write that bacteria “decide to mutate because of the antibiotic.” Random variation may already exist. Antibiotic use selects bacteria that are less susceptible, allowing them to survive and reproduce.

E. Psychotherapeutic Medicines / 精神治療藥物

Psychotherapeutic medicines (精神治療藥物) affect brain function, emotion or behaviour and must be used under medical supervision.

Type	Function	Examples
Stimulant (興奮劑)	Increases activity and alertness of the central nervous system	Amphetamine, methylphenidate
Antidepressant (抗抑鬱藥)	Treats symptoms of depression and some mood disorders	Imipramine, amitriptyline
Antipsychotic (抗精神病藥)	Treats serious psychiatric conditions involving disturbed thoughts or perception	Clozapine, haloperidol

F. Correct Use of Medicines / 正確使用藥物

- Follow the prescribed **dosage** (劑量) and schedule.
- Read the label and **instructions** carefully.
- Do not **share** prescription medicines.
- Do not use **expired** medicines.
- Inform the doctor about allergies and other medicines being taken.
- Complete a prescribed **antibiotic course**.
- Seek medical advice if serious **side effects** (副作用) occur.

G. Cosmetics / 化妝品

Cosmetics are products used **externally** to clean, protect or beautify the body. Examples include moisturisers, shampoo, perfume, lipstick, face powder, sunscreen and deodorant.

Ingredient	Function	Example
Water	Acts as a solvent or continuous phase	Purified water
Dye or pigment (染料／色素)	Provides colour	Iron oxides
Preservative (防腐劑)	Prevents microbial growth and spoilage	Permitted preservatives
Moisturiser or humectant (保濕劑)	Helps retain moisture	Glycerol
Fragrance (香料)	Provides a pleasant smell	Essential oils or fragrance compounds
Thickener (增稠劑)	Increases viscosity (黏度)	Xanthan gum
Emulsifier (乳化劑)	Forms a stable mixture of oil and water	Lecithin, emulsifying wax



Visual Link / 圖像連結: Many cosmetic products are mixtures of water, oil, fragrance, colouring and other ingredients. An **emulsifier** (乳化劑) keeps the oil and water phases mixed, while a **preservative** (防腐劑) slows microbial growth.

Cosmetic Formulation / 化妝品配方

A lotion contains oil and water. The ingredient needed to form a stable mixture is an **emulsifier**. A **moisturiser** helps retain water, while a preservative delays **spoilage**.

H. Harmful or Prohibited Ingredients / 有害或禁用成分

Some unregistered cosmetic products may contain harmful substances such as **mercury compounds** (汞化合物), hydroquinone, tretinoin or potent corticosteroids. Misuse can cause skin irritation, abnormal pigmentation, thinning of the skin, kidney damage or **nervous-system damage** (神經系統損害).

Consumers should:

- buy **registered products** from reliable sellers;
- check the ingredient list and product information;

- avoid products making **unrealistic claims** (不實宣稱) ;
- stop using a product if severe irritation occurs;
- seek professional advice when **necessary**.

I. Quick Check

1. State the function of an analgesic. [1 mark]

2. A patient with a gastric ulcer has a fever. Explain why aspirin may be unsuitable and suggest a more suitable analgesic. [3 marks]

3. Explain why antibiotics cannot treat influenza. [2 marks]

4. Explain how incomplete antibiotic treatment contributes to antibiotic resistance. [4 marks]

5. State one danger of taking too much paracetamol. [1 mark]

6. State the functions of a preservative and a moisturiser in a cosmetic cream. [2 marks]

7. Give two precautions when purchasing cosmetics. [2 marks]

5.5 Application of Nanotechnology in Industry / 奈米科技的工業應用

Suggested lesson focus: nanoscale materials, graphene, applications and risks

A. Definition / 定義

Nanotechnology is the development and application of materials, structures and devices with dimensions of approximately **1 to 100 nanometres** (1 至 100 奈米) .

$$1 \text{ nm} = 1 \times 10^{-9} \text{ m}$$

Nanomaterials may have properties different from the same material in **bulk form** (塊體形態) . This is partly because they have a very large **surface-area-to-volume ratio** (表面積與體積比) and a high proportion of atoms at the surface.

B. Graphene / 石墨烯

Graphene is a **single layer** of carbon atoms arranged in a two-dimensional **hexagonal lattice** (六角晶格) . Each carbon atom is **covalently bonded** (以共價鍵結合) to three other carbon atoms.

Important properties of graphene include:

- very high **tensile strength** (抗拉強度) ;
- **low mass**;
- flexibility and **elasticity** (彈性) ;
- high **thermal conductivity** (導熱性) ;
- high electrical conductivity and low electrical **resistance**;
- high **transparency**;
- very low **permeability** (滲透性) to gases.

C. Applications of Graphene / 石墨烯的應用

Application	Relevant Property
Flexible electronic displays	Flexible, transparent and electrically conductive
Lightweight composite materials	Strong and low in mass
Electronic components	High electrical conductivity and low resistance
Heat-management materials	High thermal conductivity
Sensitive sensors	High surface area and changes in electrical behaviour when molecules adsorb (吸附) on the surface
Protective coatings	Strong and highly impermeable (不可滲透)

D. Other Nanotechnology Applications / 其他應用

- Titanium dioxide and zinc oxide nanoparticles in sunscreens absorb or scatter **ultraviolet radiation** (紫外線) while appearing less opaque on the skin.
- Silver nanoparticles are used in some **antimicrobial coatings** (抗微生物塗層) and materials.
- Nanoparticles can act as **carriers** (載體) that deliver medicines to selected tissues.
- Nanomaterials are used in smaller electronic components and high-performance **sensors**.
- Nanocoatings can make surfaces water-resistant, scratch-resistant or **stain-resistant**.
- Nanomaterials can strengthen **composites** (複合材料) without greatly increasing mass.

E. Advantages and Concerns / 優點與疑慮

Advantages / 優點:

- improved strength, conductivity or **chemical activity**;
- smaller and **lighter** products;
- more efficient catalysts and **sensors**;
- more **targeted drug delivery** (標靶藥物輸送) ;
- reduced material usage in some applications.

Concerns / 疑慮:

- some nanoparticles may enter cells or **tissues**;
- long-term health effects may not be fully **understood**;
- nanoparticles released into the environment may be difficult to **recover**;
- careful testing, exposure control and **regulation** (監管) are required.

F. Quick Check

1. Define nanotechnology and state the nanoscale range. [2 marks]

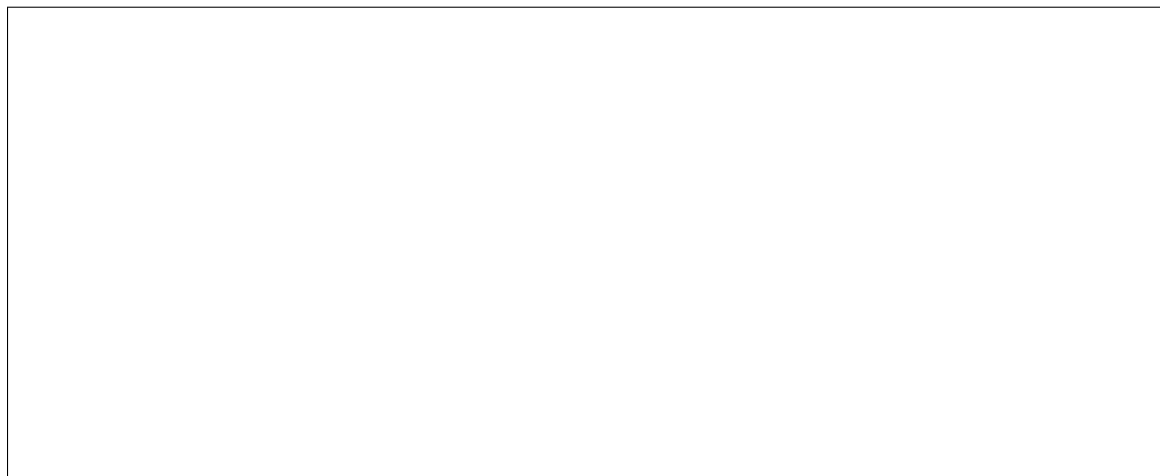
2. State four properties of graphene. [4 marks]

3. Explain why graphene is suitable for a flexible electronic display. [3 marks]

4. Explain why nanoparticles often show high chemical activity. [2 marks]

5. State one benefit and one concern associated with nanoparticles in consumer products. [2 marks]

6. Draw a simple hexagonal arrangement to represent a small section of a graphene sheet. [2 marks]



5.6 Green Technology in Industrial Waste Management / 工業廢物管理中的綠色科技

Suggested lesson focus: electrocoagulation, pollutant removal and responsible sludge handling

A. Green Technology / 綠色科技

Green Technology is the development and application of products, equipment and systems that **conserve** (保護) the environment and natural resources while reducing the negative effects of human activities.

A Green Technology process aims to:

- reduce **pollution**;
- minimise **waste**;
- conserve energy and **raw materials** (原料) ;
- reuse or recycle suitable materials;
- protect human health and **ecosystems** (生態系統) .

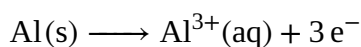
B. Industrial Wastewater / 工業廢水

Industrial wastewater may contain **suspended solids** (懸浮固體) , oils, dyes, acids, alkalis, dissolved metal ions, organic pollutants or microorganisms. It must be **treated** before discharge or reuse.

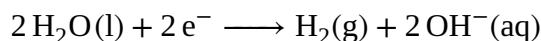
C. Electrocoagulation / 電凝聚法

Electrocoagulation is a wastewater-treatment process based on **electrolysis** (電解) . Metal electrodes, commonly aluminium or iron, are placed in the wastewater and connected to a **direct-current supply** (直流電源) .

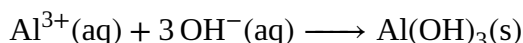
For aluminium electrodes, **oxidation** (氧化) occurs at the anode:



At the cathode, water may be **reduced** (還原) :



The aluminium ions react with hydroxide ions to form **aluminium hydroxide** (氫氧化鋁) :



The gelatinous aluminium hydroxide acts as a **coagulant** (混凝劑) . It **adsorbs** (吸附) or traps small suspended particles, dyes and some dissolved pollutants. The particles form larger **flocs** (絮凝物) that can settle, float or be filtered from the treated water.

D. Main Stages / 主要步驟

1. Electric current passes through the **wastewater**.
2. The metal anode releases **metal ions**.
3. Metal hydroxide **coagulants** (混凝劑) form in the water.
4. Pollutants are adsorbed or trapped in **flocs**.
5. Flocs settle or float and are separated as **sludge** (污泥) .
6. The treated **effluent** (處理後排放水) is tested before discharge or reuse.

E. Advantages / 優點

- removes suspended particles, dyes, oils and some **metal ions**;
- may require less added chemical **coagulant**;
- can use **compact equipment**;
- produces flocs that are relatively easy to **separate**;
- may allow treated water to be reused or discharged more **safely**.

F. Responsible Sludge Management / 負責任的污泥管理

Sludge must be **tested** before reuse or disposal. Sludge containing **toxic metals** (有毒金屬) or hazardous chemicals must not be used directly in agriculture. Only sludge shown to be safe and suitable may be processed for **beneficial use** (有益用途) .

Common Mistake

Do not automatically state that all electrocoagulation sludge can be used as fertiliser. The correct treatment depends on the pollutants collected in the sludge.

G. Quick Check

1. Define Green Technology. [2 marks]

2. State the type of reaction that occurs at the aluminium anode. [1 mark]

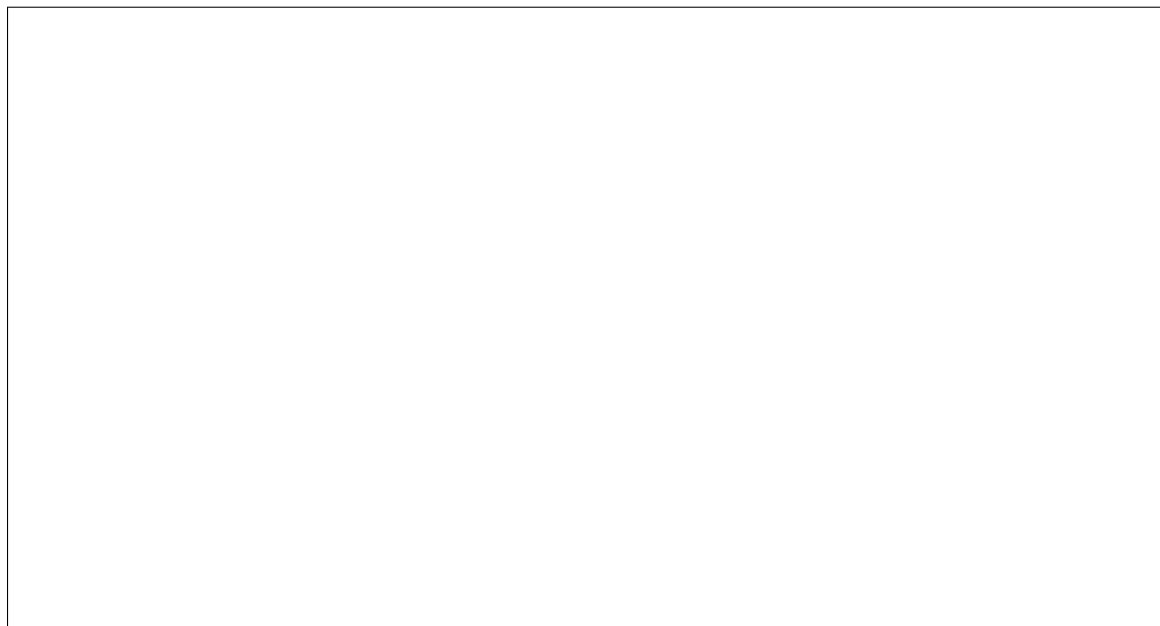
3. Write the half-equation at the aluminium anode. [2 marks]

4. Explain how aluminium hydroxide removes pollutants from wastewater. [4 marks]

5. State two advantages of electrocoagulation. [2 marks]

6. Explain why the sludge must be analysed before agricultural use. [2 marks]

7. Draw a labelled apparatus diagram for a simple electrocoagulation cell using two aluminium electrodes and a direct-current power supply. [4 marks]



Chapter Summary / 本章總結

Topic	Essential Examination Points
Oils and fats (油與脂肪)	Triglycerides ; saturated versus unsaturated chains; packing and melting point; hydrogenation using H ₂ , nickel and heat
Soap and detergent (肥皂與清潔劑)	Saponification ; salting out; hydrophilic head and hydrophobic tail; micelles ; scum in hard water; fatty acid in acidic water
Food additives (食品添加劑)	Preservative, antioxidant, flavouring, colouring, emulsifier, stabiliser and thickener ; relate each additive to the food product
Medicines (藥物)	Analgesics, antibiotics and psychotherapeutic medicines; correct dosage; side effects; antibiotic resistance
Cosmetics (化妝品)	Basic ingredients and functions; harmful unregistered ingredients; safe consumer choices
Nanotechnology (奈米科技)	1–100 nm ; high surface-area-to-volume ratio; graphene properties and applications; benefits and concerns
Green Technology (綠色科技)	Electrocoagulation ; electrode reactions; hydroxide flocs; pollutant removal; testing and safe handling of sludge

Practice Set A: Multiple-Choice Questions / 選擇題

Choose the best answer. (選出最佳答案。)

- Which substances react to form a triglyceride (三酸甘油酯)?
 - Glycerol and three fatty acid molecules
 - Glycerol and sodium hydroxide
 - An alkane and oxygen
 - Three alcohol molecules and one ester
- Which statement describes a saturated fatty acid (飽和脂肪酸)?
 - It contains at least one carbon-carbon double bond.
 - It contains only carbon-carbon single bonds in its carbon chain.
 - It must be obtained from an animal.
 - It always exists as a gas at room temperature.
- Why do many unsaturated oil (不飽和油)s have lower melting points than saturated fats?
 - Their molecules are ionic.
 - Their double bonds produce bends that reduce close packing.
 - They contain no carbon atoms.
 - Their ester bonds are completely broken.
- Which conditions are used to hydrogenate vegetable oil?
 - H₂, nickel and heat
 - O₂, iron and cooling
 - N₂, sodium hydroxide and heat
 - Steam, copper and room temperature
- Which change occurs during hydrogenation (氫化反應)?
 - Carbon-carbon single bonds become carbon-carbon double bonds.
 - Carbon-carbon double bonds become carbon-carbon single bonds.
 - Ester groups become alcohol groups by hydrolysis.
 - Fatty acids are converted into glycerol only.
- Which substance is produced together with soap during saponification (皂化反應)?
 - Glycerol
 - Ethene
 - Sodium chloride
 - Hydrogen
- Why is saturated sodium chloride solution added during soap preparation?
 - To convert soap into detergent

- B. To increase the solubility of soap
 - C. To precipitate soap from the mixture
 - D. To oxidise glycerol
8. Which part of a soap anion dissolves in grease?
- A. The hydrophobic hydrocarbon tail (疏水性碳氫尾部)
 - B. The hydrophilic ionic head (親水性離子頭部)
 - C. The sodium ion only
 - D. The oxygen atom only
9. Why is soap less effective in hard water (硬水) ?
- A. Soap evaporates rapidly.
 - B. Soap forms insoluble salts with calcium and magnesium ions.
 - C. Soap reacts with water to form oxygen.
 - D. Soap increases the surface tension of water.
10. Which cleaning agent is generally more effective in hard water?
- A. Detergent
 - B. Soap
 - C. Vegetable oil
 - D. Glycerol
11. Which additive slows the oxidation of fats and oils?
- A. Antioxidant (抗氧化劑)
 - B. Thickener
 - C. Colouring agent
 - D. Flavouring agent
12. Which substance is commonly used as a thickener in jam?
- A. Pectin
 - B. Sodium chloride
 - C. Ascorbic acid
 - D. Monosodium glutamate
13. Lecithin is added to mayonnaise mainly to:
- A. preserve it by killing all microorganisms;
 - B. keep the oil and water phases mixed;
 - C. make it strongly acidic;
 - D. produce carbon dioxide.
14. Which statement about antibiotics (抗生素) is correct?

- A. They treat all viral infections.
 - B. They kill bacteria or inhibit bacterial growth.
 - C. They should be shared with family members.
 - D. They should be stopped as soon as symptoms improve.
15. Excessive intake of which analgesic (止痛藥) can cause severe liver damage?
- A. Paracetamol
 - B. Penicillin
 - C. Lecithin
 - D. Sodium benzoate
16. What is the main function of a moisturiser in cosmetics?
- A. To retain moisture
 - B. To increase microbial growth
 - C. To produce metal ions
 - D. To remove all pigments
17. Which substance may be harmful if illegally added to skin-whitening cosmetics?
- A. Mercury compound
 - B. Pectin
 - C. Glycerol
 - D. Sodium chloride
18. Which range represents nanoscale dimensions?
- A. 1 to 100 m
 - B. 1 to 100 cm
 - C. 1 to 100 μm
 - D. 1 to 100 nm
19. Which combination makes graphene (石墨烯) suitable for flexible electronic displays?
- A. Flexible, transparent and electrically conductive
 - B. Opaque, brittle and insulating
 - C. Soluble, soft and non-conductive
 - D. Dense, magnetic and radioactive
20. What is the main purpose of electrocoagulation (電凝聚法) ?
- A. To convert oils into margarine
 - B. To remove pollutants from wastewater
 - C. To manufacture antibiotics
 - D. To add flavouring to food
21. Which process occurs at the aluminium anode during electrocoagulation?

- A. Reduction
- B. Neutralisation
- C. Oxidation
- D. Esterification

22. What is the role of $\text{Al}(\text{OH})_3$ in electrocoagulation?

- A. It acts as a coagulant (混凝劑) that traps pollutants in flocs.
- B. It acts as a food preservative.
- C. It converts grease into soap.
- D. It increases the hardness of water.

23. Which statement about electrocoagulation sludge (污泥) is correct?

- A. It is always safe to use as fertiliser.
- B. It must be analysed before reuse or disposal.
- C. It contains only pure water.
- D. It should always be discharged directly into rivers.

24. A student stops taking antibiotics after one day. Which outcome is most likely?

- A. All viruses are destroyed.
- B. Surviving less-susceptible bacteria may reproduce.
- C. The antibiotic becomes a detergent.
- D. The medicine becomes more concentrated in the packet.

25. Which statement best explains the high activity of many nanoparticles?

- A. They have a high surface-area-to-volume ratio (表面積與體積比).
- B. They have no atoms at their surface.
- C. They are always electrically neutral.
- D. They cannot interact with other substances.

MCQ Response Grid / 選擇題作答表

1.	2.	3.	4.	5.
6.	7.	8.	9.	10.
11.	12.	13.	14.	15.
16.	17.	18.	19.	20.
21.	22.	23.	24.	25.

Practice Set B: Structured Questions / 結構題

Question 1: Oils, Fats and Hydrogenation

[12 marks]

Sample P is a liquid vegetable oil. Sample Q is a solid animal fat.

- (a) State the main type of fatty acid present in P and Q.

[2 marks]

- (b) Explain why P is liquid while Q is solid at room temperature.

[4 marks]

- (c) Describe how P can be converted into a semi-solid product. Include the reagent, catalyst, temperature and structural change.

[4 marks]

- (d) State the type of reaction in part (c).

[1 mark]

- (e) State one use of the semi-solid product formed.

[1 mark]

Question 2: Preparation of Soap**[12 marks]**

A student heats palm oil with concentrated sodium hydroxide solution and later adds saturated sodium chloride solution.

- (a) Name the reaction. [1 mark]

- (b) State the two products formed before sodium chloride is added. [2 marks]

- (c) Write a general equation for the reaction. [3 marks]

- (d) Explain the function of sodium chloride. [2 marks]

- (e) State two steps used to separate and purify the soap after cooling. [2 marks]

- (f) State one safety precaution during the experiment and explain its importance. [2 marks]

Question 3: Cleansing Action and Hard Water**[14 marks]**

A greasy cloth is washed using detergent in hard water.

(a) Name the two parts of a detergent anion.

[2 marks]

(b) State which part is attracted to grease and which part is attracted to water.

[2 marks]

(c) Explain the cleansing action of the detergent.

[6 marks]

(d) Explain why detergent is more effective than soap in hard water.

[4 marks]

Question 4: Food Additives**[12 marks]**

A bottle of salad dressing lists the following ingredients: lecithin, ascorbic acid, sodium benzoate, colouring and flavouring.

(a) State the category and function of lecithin.

[2 marks]

(b) State the category and function of ascorbic acid.

[2 marks]

(c) State the category and function of sodium benzoate.

[2 marks]

(d) Explain why both an emulsifier (乳化劑) and an antioxidant are useful in this food product. [4 marks]

(e) State one reason why food additives must be used within permitted limits.

[2 marks]

Question 5: Medicines**[14 marks]**

A patient with a gastric ulcer has a bacterial throat infection and a fever.

- (a) State the function of an analgesic. [1 mark]

- (b) Explain why aspirin may be unsuitable for the patient. [2 marks]

- (c) Suggest a more suitable analgesic and state one danger of overdose. [2 marks]

- (d) State the type of medicine used to treat the bacterial infection. [1 mark]

- (e) Explain why this medicine would not treat influenza. [2 marks]

- (f) Explain why the patient must complete the prescribed course. [4 marks]

(g) State two rules for the safe use of medicines.

[2 marks]

Question 6: Cosmetics and Consumer Safety

[12 marks]

A moisturising cream contains water, glycerol, an emulsifier, fragrance and a preservative.

(a) State the function of water.

[1 mark]

(b) State the function of glycerol.

[1 mark]

(c) Explain why an emulsifier is needed in the cream.

[2 marks]

(d) Explain why a preservative is needed.

[2 marks]

(e) Name two harmful substances that may be present in unregistered cosmetics.

[2 marks]

(f) Explain two possible harmful effects of one substance named in part (e).

[2 marks]

(g) Suggest two precautions when purchasing cosmetics.

[2 marks]

Question 7: Nanotechnology and Graphene

[12 marks]

(a) Define nanotechnology (奈米科技) .

[2 marks]

(b) State four properties of graphene.

[4 marks]

(c) Explain why graphene is suitable for flexible electronic displays.

[3 marks]

(d) State one application of a nanoparticle other than graphene and relate the application to a useful property.

[2 marks]

(e) State one possible concern associated with nanoparticles.

[1 mark]

Question 8: Electrocoagulation

[14 marks]

Industrial wastewater containing suspended particles, dyes and metal ions is treated using aluminium electrodes.

(a) Name the treatment process.

[1 mark]

(b) State the type of reaction at the aluminium anode.

[1 mark]

(c) Write the half-equation at the aluminium anode.

[2 marks]

(d) Write a possible half-equation at the cathode.

[2 marks]

(e) Explain how aluminium hydroxide helps remove pollutants.

[4 marks]

(f) State two advantages of electrocoagulation.

[2 marks]

(g) Explain why the sludge must be tested before reuse or disposal.

[2 marks]

Practice Set C: Higher-Order Thinking Questions / 高階思維題

Question 1: Selecting a Cleaning Agent

[8 marks]

A household is supplied with hard water. Two products are available: a readily biodegradable soap and a detergent that remains effective in hard water.

Compare the products and recommend the more suitable cleaning agent for washing clothes. Your answer must include cleansing effectiveness and environmental considerations.

Question 2: Reformulating a Food Product**[8 marks]**

A salad dressing separates into an oil layer and a water layer. After long storage, the oil develops an unpleasant smell.

Suggest two suitable food additives and explain how each additive solves one of the problems.

[10 marks]

Evaluate the student's actions and explain the correct steps that should have been taken.

[illegible]

[10 marks]

Propose a treatment strategy based on electrocoagulation. Explain the electrode process, formation of flocs, separation of sludge and environmental benefits.

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Practice Set D: Experimental Design / 實驗設計

Experiment 1: Comparing Soap and Detergent

Design an experiment to compare the effectiveness of soap and detergent in soft water and hard water.

Problem statement

Hypothesis

Variables

- Manipulated variable:

- Responding variable:

- Controlled variables:

Apparatus and materials

Procedure

Table for observations

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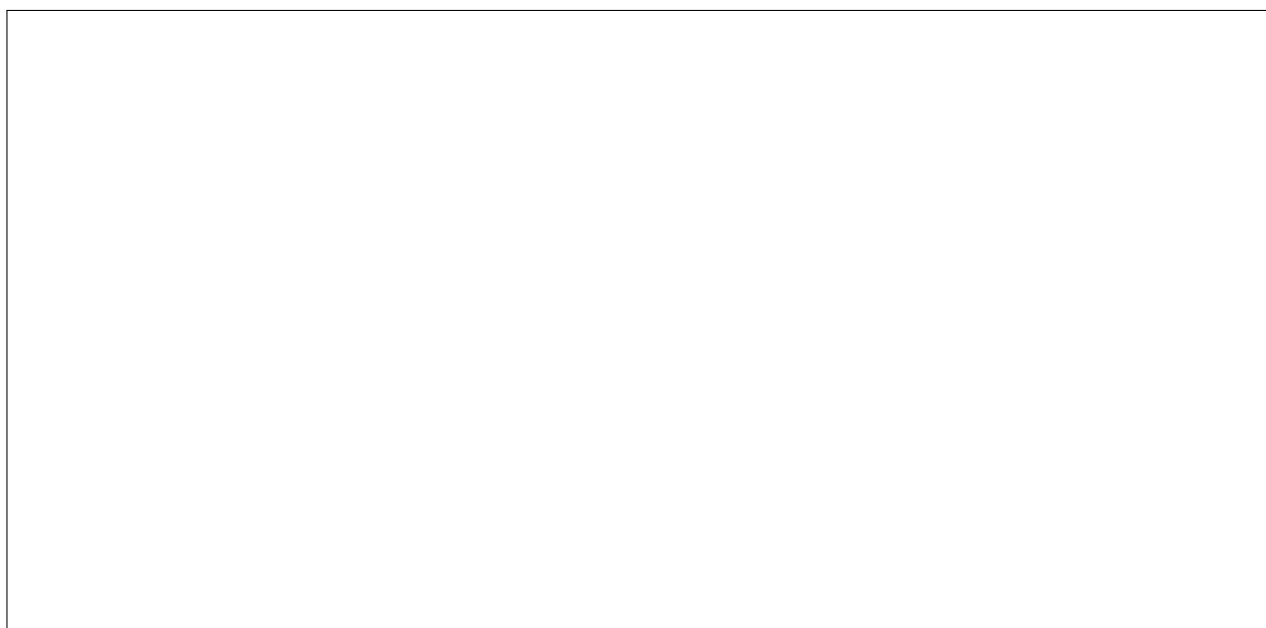
Expected conclusion

Describe a laboratory experiment to prepare soap from vegetable oil. Include the apparatus and materials, procedure, purpose of sodium hydroxide, purpose of sodium chloride, separation method, observation and safety precautions.

[illegible]

Functions of sodium hydroxide and sodium chloride

Safety precautions

Experimental setup

Final Self-Assessment / 最後自我評估

Tick the box that best represents your current understanding.

Skill	Confident	Need Practice	Need Help
I can explain why oils and fats have different melting points.	☐	☐	☐
I can explain hydrogenation using bonds and molecular packing.	☐	☐	☐
I can explain saponification and salting out.	☐	☐	☐
I can describe micelle formation and cleansing action.	☐	☐	☐
I can explain why soap is ineffective in hard or acidic water.	☐	☐	☐
I can identify food additives from a food label.	☐	☐	☐
I can explain correct medicine use and antibiotic resistance (抗生素抗藥性) .	☐	☐	☐
I can relate graphene properties to applications.	☐	☐	☐
I can explain electrocoagulation and sludge management.	☐	☐	☐

Teacher Answer Scheme

Quick Check: 5.1 Oils and Fats

1. Esters or triglycerides.
2. Glycerol or propan-1,2,3-triol.
3. Saturated fatty acid: only carbon-carbon single bonds. Unsaturated fatty acid: at least one carbon-carbon double bond.
4. Double bonds produce bends in the hydrocarbon chains; molecules pack less closely; intermolecular attractions are weaker; melting point is lower.
5. Hydrogen gas; nickel catalyst; about 180°C.
6. The drawing should show $-\text{CH}=\text{CH}- + \text{H}_2 \longrightarrow -\text{CH}_2-\text{CH}_2-$.

Quick Check: 5.2 Cleaning Agents

1. Alkaline hydrolysis of an oil or fat to form soap and glycerol.
2. Soap and glycerol.
3. Sodium chloride reduces the solubility of soap and precipitates it; this is salting out.
4. Hydrophobic hydrocarbon tail attracted to grease; hydrophilic ionic head attracted to water.
5. Micelle: tails point inward into grease; ionic heads point outward into water.
6. $2 \text{RCOO}^- + \text{Ca}^{2+} \longrightarrow (\text{RCOO})_2\text{Ca}(\text{s})$.
7. Soap anions react with H^+ to form insoluble fatty acids, RCOOH , reducing cleansing effectiveness.

Quick Check: 5.3 Food Additives

1. Lecithin: emulsifier/stabiliser; keeps oil and water mixed.
2. Ascorbic acid: antioxidant; slows oxidation and rancidity.
3. Sodium benzoate: preservative; inhibits microbial growth.
4. It prevents the oil and water phases from separating and produces a stable emulsion.
5. Pectin is a thickener; it absorbs water and forms a gel, increasing viscosity.
6. To reduce harmful health effects and ensure safe consumption.

Quick Check: 5.4 Medicines and Cosmetics

1. Relieves pain without necessarily curing the underlying disease.
2. Aspirin may irritate the stomach lining or worsen bleeding; paracetamol is generally more suitable for fever in this case.
3. Antibiotics act against bacteria, whereas influenza is caused by a virus.

4. Early stopping leaves some bacteria alive; less-susceptible bacteria survive and reproduce; the proportion of resistant bacteria increases; infection becomes harder to treat.
5. Severe liver damage.
6. Preservative prevents microbial growth; moisturiser/humectant helps retain moisture.
7. Any two: buy registered products; use reliable sellers; check ingredients; avoid unrealistic claims; stop use if severe irritation occurs.

Quick Check: 5.5 Nanotechnology

1. Development and application of materials or devices approximately 1 to 100 nm in size.
2. Any four: strong, light, flexible, transparent, electrically conductive, thermally conductive, low resistance, impermeable.
3. Graphene is flexible, transparent and electrically conductive.
4. Nanoparticles have a high surface-area-to-volume ratio and a large proportion of surface atoms available for interaction.
5. Benefit: improved performance, targeted delivery, lighter materials, etc. Concern: uncertain toxicity, entry into cells, environmental persistence, etc.
6. A two-dimensional hexagonal lattice is expected.

Quick Check: 5.6 Green Technology

1. Development and application of products, equipment or systems that conserve the environment and natural resources while reducing negative impacts.
2. Oxidation.
3. $\text{Al(s)} \longrightarrow \text{Al}^{3+}(\text{aq}) + 3\text{e}^{-}$.
4. Al(OH)_3 forms gelatinous flocs that adsorb or trap small particles, dyes and some ions; larger flocs settle, float or are filtered.
5. Any two suitable advantages from the notes.
6. Sludge may contain toxic metals or hazardous chemicals and must be tested before reuse or disposal.
7. Diagram should show two aluminium electrodes connected to a DC supply and immersed in wastewater.

Practice Set A: Multiple-Choice Answers

1 A	2 B	3 B	4 A	5 B
6 A	7 C	8 A	9 B	10 A
11 A	12 A	13 B	14 B	15 A
16 A	17 A	18 D	19 A	20 B
21 C	22 A	23 B	24 B	25 A

Practice Set B: Structured Questions / 結構題

Question 1

- P: mainly unsaturated fatty acids; Q: mainly saturated fatty acids.
- P contains bent chains caused by double bonds, packs less closely and has weaker intermolecular attractions; Q has straighter chains, closer packing and stronger attractions.
- React P with hydrogen gas using nickel at about 180°C; hydrogen adds across C=C bonds, increasing saturation and melting point.
- Addition reaction or hydrogenation.
- Margarine or another acceptable semi-solid fat product.

Question 2

- Saponification.
- Soap and glycerol.
- Triglyceride + 3 NaOH $\longrightarrow$ Glycerol + 3 RCOONa.
- Sodium chloride lowers soap solubility and precipitates the soap.
- Filter; wash with cold distilled water; press/dry.
- Suitable precaution with reason, for example wear eye protection because concentrated sodium hydroxide is corrosive.

Question 3

- Hydrophobic hydrocarbon tail; hydrophilic ionic head.
- Tail enters grease; head remains in water.
- Lower surface tension; tails enter grease; heads remain in water; agitation breaks grease; micelles form; charged heads prevent recombination; rinsing removes droplets.
- Hard water contains Ca^{2+} and Mg^{2+} ; soap forms insoluble scum; detergent does not normally form scum and remains effective.

Question 4

- Lecithin: emulsifier/stabiliser; keeps phases mixed.
- Ascorbic acid: antioxidant; slows oxidation.
- Sodium benzoate: preservative; inhibits microbial growth.
- Emulsifier prevents oil-water separation; antioxidant delays oxidation and rancidity.
- Excessive intake may cause harmful or sensitivity reactions; permitted limits protect consumers.

Question 5

- Relieves pain.
- Aspirin may irritate the stomach lining or worsen gastric bleeding.
- Paracetamol; overdose can cause severe liver damage.
- Antibiotic.
- Influenza is caused by a virus, not bacteria.

- Complete resistance explanation as given above.
- Any two accepted medicine-safety rules.

Question 6

- Water: solvent or continuous phase.
- Glycerol: moisturiser/humectant that retains moisture.
- Emulsifier keeps oil and water phases uniformly mixed.
- Preservative inhibits microbial growth and spoilage.
- Any two: mercury compounds, hydroquinone, tretinoin, potent corticosteroids.
- Appropriate harmful effects linked to the selected substance.
- Any two accepted precautions.

Question 7

- Nanotechnology definition including 1–100 nm.
- Four valid graphene properties.
- Flexible, transparent and electrically conductive.
- Example with property link, such as TiO_2 nanoparticles in sunscreen because they absorb/scatter UV radiation while appearing less opaque.
- One valid health or environmental concern.

Question 8

- Electrocoagulation.
- Oxidation.
- $\text{Al} \longrightarrow \text{Al}^{3+} + 3\text{e}^-$.
- $2\text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{H}_2 + 2\text{OH}^-$.
- Formation of $\text{Al}(\text{OH})_3$ flocs; adsorption/trapping of pollutants; formation of larger aggregates; settling/flotation/filtration.
- Any two suitable advantages.
- Sludge may contain toxic pollutants and must be characterised before safe reuse or disposal.

Practice Set C: Suggested Marking Points

Question 1: Soap is biodegradable but forms scum in hard water and cleans less effectively. Detergent remains effective in hard water but environmental impact depends on biodegradability and formulation. Recommendation should be justified; a biodegradable detergent suitable for hard water would be ideal if available.

Question 2: Lecithin or another emulsifier prevents separation; ascorbic acid or another antioxidant delays oxidation and rancidity.

Question 3: The student should not self-medicate with another person's prescription. Diagnosis is needed because sore throat may be bacterial or viral. Correct medicine and dose must be prescribed. Complete the course if an antibiotic is prescribed. Do not keep or share leftovers.

Question 4: Aluminium electrodes and DC supply; anode oxidation; Al^{3+} and OH^- form $\text{Al}(\text{OH})_3$; flocs trap pollutants; separate sludge; test effluent and sludge; safely reuse water or discharge only when standards are met.

Practice Set D: Experimental Design / 實驗設計 Guidance

Experiment 1: Use equal volumes and concentrations of soap/detergent solutions and equal volumes of soft/hard water. Shake each test tube using the same number and strength of shakes. Measure lather height or time for lather to disappear. Detergent should produce more lather than soap in hard water.

Experiment 2: Heat oil with concentrated sodium hydroxide while stirring; add saturated sodium chloride; cool; filter; wash with cold water; press and dry. Wear goggles and gloves because sodium hydroxide is corrosive; use a water bath or controlled heating and avoid direct splashing.